

30. Morphological Operations Using OpenCV – Dilation Technique

Aim

To perform **morphological dilation** on an image using OpenCV.

Procedure

1. Read the input image using OpenCV.
2. Convert the image into grayscale.
3. Create a structuring element (kernel).
4. Apply the **dilation** operation using cv2.dilate().
5. Display the original and dilated images.
6. Save the output image.

Program

```
import cv2

import numpy as np

# Read input image
img = cv2.imread("input.jpg")

# Check image
if img is None:
    print("Error: Image not found")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create structuring element
kernel = np.ones((5, 5), np.uint8)
```

```
# Apply dilation
```

```
dilated = cv2.dilate(gray, kernel, iterations=1)
```

```
# Display images
```

```
cv2.imshow("Original Image", gray)
```

```
cv2.imshow("Dilated Image", dilated)
```

```
# Save output
```

```
cv2.imwrite("dilated_image.jpg", dilated)
```

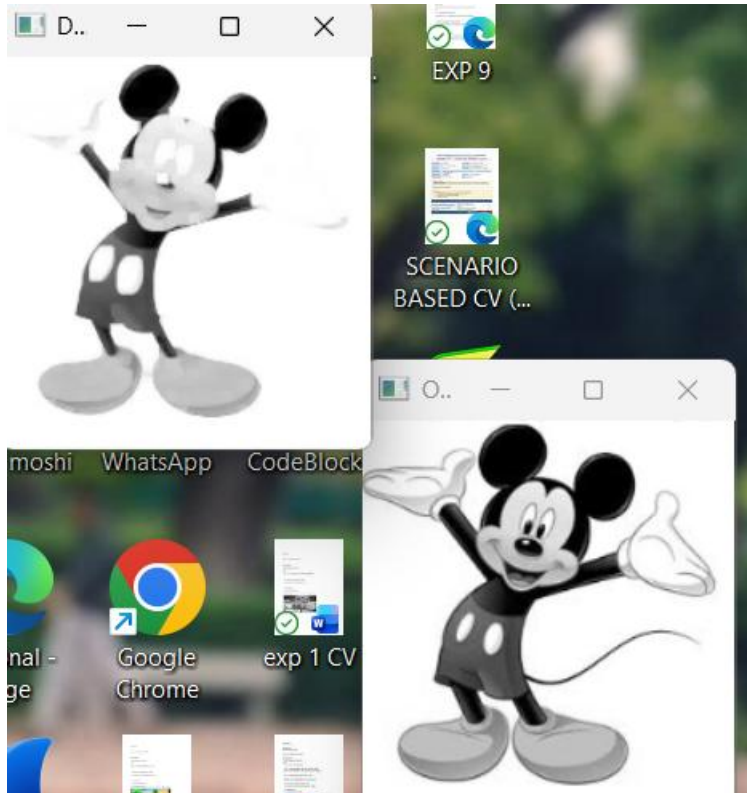
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

The **dilation operation was successfully performed using OpenCV**. The objects in the image were expanded and small gaps or holes were reduced using the structuring element.